

```

import numpy as np

arr = np.array([1, 2, 3, 4, 5]) # 1D array
arr2 = np.array([[1, 2, 3], [4, 5, 6]]) # 2D array
arr3 = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]]) # 3D array

# Elements in the array
print(arr.size)
print(arr2.size)
print(arr3.size)

print("\n")
# Dimensions of the array
print(arr.ndim)
print(arr2.ndim)
print(arr3.ndim)

5
6
8

1
2
3

print("Shape", arr.shape)

print("Reshaping the array")
arr.reshape(5, 1)

Shape (5,)
Reshaping the array

array([[1],
       [2],
       [3],
       [4],
       [5]])

arr_copy = arr.copy()
print("Original array:", arr)
print("Copied array:", arr_copy)

Original array: [1 2 3 4 5]
Copied array: [1 2 3 4 5]

demo = arr2.ravel()
print("Ravelled array:", demo)
demo[0] = 10
print("Modified ravelled array:", demo)

```

Ravelled array: [1 2 3 4 5 6]

Modified ravelled array: [10 2 3 4 5 6]

```
print(arr.sum()) # Sum of all elements in the array
print(arr2.sum(axis=1)) # Sum of each row in the array
```

```
15
[15 15]
```

#Broadcasting

```
print(arr*3) # Each element will be multiplied by 3
```

```
[ 3  6  9 12 15]
```

Masking

```
print(arr[arr > 2]) # Elements greater > 2
```

```
[3 4 5]
```

Where

```
print(np.where(arr > 2)) # positions of elements greater than 2
```

```
(array([2, 3, 4]),)
```

```
marks = np.array([67, 45, 23, 76, 98, 45, 98, 44, 90, 95, 34, 67, 65,
34, 89, 79, 87, 80, 76])
```

```
grades = np.select(
    [marks >= 90, marks >= 70, marks >= 50, marks >= 33],
    ["A", "B", "C", "D"],
    default="F"
)
```

```
print("Grades for the marks are:")
for mark, grade in zip(marks, grades):
    print(f"Mark: {mark} - Grade: {grade}")
```

Grades for the marks are:

Mark: 67 - Grade: C

Mark: 45 - Grade: D

Mark: 23 - Grade: F

Mark: 76 - Grade: B

Mark: 98 - Grade: A

Mark: 45 - Grade: D

Mark: 98 - Grade: A

Mark: 44 - Grade: D

Mark: 90 - Grade: A

Mark: 95 - Grade: A

Mark: 34 - Grade: D

Mark: 67 - Grade: C

Mark: 65 - Grade: C
Mark: 34 - Grade: D
Mark: 89 - Grade: B
Mark: 79 - Grade: B
Mark: 87 - Grade: B
Mark: 80 - Grade: B
Mark: 76 - Grade: B

```
import pandas as pd

data = {
    'Name': ['Vaibhav', 'Aryan', 'Rudra', 'Riyansh', 'XYZ'],
    'Age': [1, 2, 99, 50, 22],
    'City': ['New York', 'Los Angeles', 'Chicago', 'Houston',
'Phoenix']
}

df = pd.DataFrame(data)
print(df)
```

	Name	Age	City
0	Vaibhav	1	New York
1	Aryan	2	Los Angeles
2	Rudra	99	Chicago
3	Riyansh	50	Houston
4	XYZ	22	Phoenix

```
print("first two rows of the dataframe are: ")
print(df.head(2))
print('\n')
print("last two rows of the dataframe are: ")
print(df.tail(2))
```

first two rows of the dataframe are:

	Name	Age	City
0	Vaibhav	1	New York
1	Aryan	2	Los Angeles

last two rows of the dataframe are:

	Name	Age	City
3	Riyansh	50	Houston
4	XYZ	22	Phoenix

```
print("eleminating a column from the dataframe: ")
print(df.drop("City", axis=1))
print("\n Eleminating a row from the dataframe: ")
print(df.drop(index=4, axis=0))
```

eleminating a column from the dataframe:

	Name	Age
0	Vaibhav	1
1	Aryan	2
2	Rudra	99
3	Riyansh	50
4	XYZ	22

Eleminating a row from the dataframe:

	Name	Age	City
--	------	-----	------

0	Vaibhav	1	New York
1	Aryan	2	Los Angeles
2	Rudra	99	Chicago
3	Riyansh	50	Houston

```
print(df.loc[0])
```

```
Name      Vaibhav
Age        1
City      New York
Name: 0, dtype: object
```

```
marks = pd.DataFrame({
    'Marks' : [-100, 100, 0, 50, 90]
})
```

```
df = pd.merge(df, marks, left_index=True, right_index=True)
print(df)
```

	Name	Age	City	Marks
0	Vaibhav	1	New York	-100
1	Aryan	2	Los Angeles	100
2	Rudra	99	Chicago	0
3	Riyansh	50	Houston	50
4	XYZ	22	Phoenix	90

```
print("Description of the DataFrame:", df.describe())
```

Description of the DataFrame:		Age	Marks
count	5.000000	5.000000	
mean	34.800000	28.000000	
std	41.032914	81.670068	
min	1.000000	-100.000000	
25%	2.000000	0.000000	
50%	22.000000	50.000000	
75%	50.000000	90.000000	
max	99.000000	100.000000	

```
print("Check for null values: ", df.isnull().sum())
```

```
Check for null values:  Name      0
Age      0
City      0
Marks     0
dtype: int64
```

```
df['Grade'] = df['Marks'].apply(lambda x: 'A' if x < 0 else ('B' if x
>= 100 else ('C' if x == 0 else ('D' if x == 50 else 'F'))))
print(df)
```

	Name	Age	City	Marks	Grade
0	Vaibhav	1	New York	-100	A

1	Aryan	2	Los Angeles	100	B
2	Rudra	99	Chicago	0	C
3	Riyansh	50	Houston	50	D
4	XYZ	22	Phoenix	90	F

```
df1 = pd.DataFrame({'a': [None, 3, 3], 'b' : [1, 2, None]})
# Fill null values with given 0 and inplace = True means, change in
the original dataframe.
df1.fillna(0, inplace=True)
print(df1)
```

	a	b
0	0.0	1.0
1	3.0	2.0
2	3.0	0.0

```
df2 = pd.DataFrame({
    'Name': ['Vaibhav', 'Aryan', 'Rudra', 'Riyansh', 'Riyansh']
})
```

```
print("Drop duplicates from the DataFrame:")
print(df2.drop_duplicates())
```

Drop duplicates from the DataFrame:

	Name
0	Vaibhav
1	Aryan
2	Rudra
3	Riyansh

```
print("Rename columns:\n",df['Name'].rename('Full Name'))
```

Rename columns:

0	Vaibhav
1	Aryan
2	Rudra
3	Riyansh
4	XYZ

Name: Full Name, dtype: str

```
print(df['Marks'].mean())
print(df['Age'].mean())
```

28.0
34.8

```
df2 = pd.DataFrame({
    'Name': ['Vaibhav', 'Aryan', 'Rudra', 'Riyansh', 'XYZ'],
    'Groups': ['A', 'B', 'A', 'B', 'C'],
})
```

```
print(df2.groupby('Groups')['Name'].count())
```

Groups

A 2

B 2

C 1

Name: Name, dtype: int64

```
# Saving the DataFrame into a CSV file
```

```
df.to_csv('output.csv', index=False)
```

```
# Loading the CSV file into a DataFrame
```

```
df_loaded = pd.read_csv('output.csv')
```

```
print("Loaded DataFrame from CSV:\n", df_loaded)
```

Loaded DataFrame from CSV:

	Name	Age	City
0	Vaibhav	1	New York
1	Aryan	2	Los Angeles
2	Rudra	99	Chicago
3	Riyansh	50	Houston
4	XYZ	22	Phoenix